

谷 英典

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ORCID	Google Scholar	GitHub	Publications	Data & Code
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62 査読付き論文	2,323 総被引用	23 H-INDEX	34 I10-INDEX
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出典：Google Scholar（2026-07-20 時点）／論文数は ORCID

研究概要

専門は分子生物学、特に**長鎖ノンコーディングRNA (lncRNA)**の機能と代謝。細胞内RNAの安定性（半減期）を網羅的に測定する**BRIC-seq**法を開発し、RNA分解を機能の一側面として捉える研究を展開している。近年は単一細胞でのRNA turnover、RNA創薬、RNA生物学への計算・AI手法の応用にも取り組む。

研究分野

- 長鎖ノンコーディングRNA (lncRNA) の機能と代謝
- RNA安定性・半減期のゲノムワイド計測 (BRIC-seq)
- 転写後制御・RNA分解経路
- 単一細胞でのRNA turnover
- RNA創薬、およびRNA生物学への計算・AI手法の応用

職歴

2023 -	准教授 横浜薬科大学 薬学部 健康薬学科 Associate Professor, Yokohama University of Pharmacy
2012 - 2023	研究員 → 主任研究員 産業技術総合研究所 (AIST) Researcher / Senior Researcher, AIST
2009 - 2012	博士研究員 (ポスドク) 東京大学 Postdoctoral Researcher, University of Tokyo

学歴

2008	博士 (工学) 早稲田大学大学院 理工学研究科 Ph.D. in Engineering, Waseda University 早稲田大学 理工学部 応用化学科 (学部) B.Eng., Applied Chemistry, Waseda University
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BRIC-seq (5′ -bromo-uridine immunoprecipitation chase sequencing) を開発し、細胞内RNAの安定性（半減期）をゲノムワイドに測定する手法を確立。数百の短寿命ノンコーディング転写産物を同定した（Genome Research, 2012）。以後、RNA安定性を機能の指標として捉える研究を一貫して展開している。

主要業績

1. **Tani H.**, Mizutani R., Salam K.A., Tano K., Ijiri K., Wakamatsu A., et al. Genome-wide determination of RNA stability reveals hundreds of short-lived noncoding transcripts in mammals. *Genome Research* 22 (2012). [doi:10.1101/gr.130559.111](https://doi.org/10.1101/gr.130559.111)
2. **Tani H.**, Imamachi N., Salam K.A., Mizutani R., Ijiri K., Irie T., et al. Identification of hundreds of novel UPF1 target transcripts by direct determination of whole transcriptome stability. *RNA Biology* 9 (2012). [doi:10.4161/rna.22360](https://doi.org/10.4161/rna.22360)
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5. Imamachi N., **Tani H.**, Mizutani R., Imamura K., Irie T., Suzuki Y., et al. BRIC-seq: A genome-wide approach for determining RNA stability in mammalian cells. *Methods* 67 (2014). [doi:10.1016/j.ymeth.2013.07.014](https://doi.org/10.1016/j.ymeth.2013.07.014)
6. **Tani H.**, Onuma Y., Ito Y., Torimura M. Long Non-Coding RNAs as Surrogate Indicators for Chemical Stress Responses in Human-Induced Pluripotent Stem Cells. *PLoS ONE* 9 (2014). [doi:10.1371/journal.pone.0106282](https://doi.org/10.1371/journal.pone.0106282)
7. **Tani H.**, Torimura M. Identification of short-lived long non-coding RNAs as surrogate indicators for chemical stress response. *Biochemical and Biophysical Research Communications* 439 (2013). [doi:10.1016/j.bbrc.2013.09.006](https://doi.org/10.1016/j.bbrc.2013.09.006)
8. **Tani H.** Recent Advances and Prospects in RNA Drug Development. *International Journal of Molecular Sciences* 25 (2024). [doi:10.3390/ijms252212284](https://doi.org/10.3390/ijms252212284)
9. **Tani H.**, Teramura T., Adachi K., Tsuneda S., Kurata S., Nakamura K., et al. Technique for Quantitative Detection of Specific DNA Sequences Using Alternately Binding Quenching Probe Competitive Assay Combined with Loop-Mediated Isothermal Amplification. *Analytical Chemistry* 79 (2007). [doi:10.1021/ac070041e](https://doi.org/10.1021/ac070041e)
10. Mizutani R., Wakamatsu A., Tanaka N., Yoshida H., Tochigi N., Suzuki Y., et al. Identification and Characterization of Novel Genotoxic Stress-Inducible Nuclear Long Noncoding RNAs in Mammalian Cells. *PLoS ONE* 7 (2012). [doi:10.1371/journal.pone.0034949](https://doi.org/10.1371/journal.pone.0034949)
11. Maekawa S., Imamachi N., Irie T., **Tani H.**, Matsumoto K., Mizutani R., et al. Analysis of RNA decay factor mediated RNA stability contributions on RNA abundance. *BMC Genomics* 16 (2015). [doi:10.1186/s12864-015-1358-y](https://doi.org/10.1186/s12864-015-1358-y)
12. **Tani H.**, Miyata R., Ichikawa K., Morishita S., Kurata S., Nakamura K., et al. Universal Quenching Probe System: Flexible, Specific, and Cost-Effective Real-Time Polymerase Chain Reaction Method. *Analytical Chemistry* 81 (2009). [doi:10.1021/ac900414u](https://doi.org/10.1021/ac900414u)

教育

横浜薬科大学 担当科目

- 情報科学入門（1年次・必修）
- 情報処理演習（1年次・必修）
- 生物系薬学演習1（4年次・必修）
- 生物系実習II（2年次・必修／科目責任者）

学会活動・査読

- 査読：*Microbiology and Molecular Biology Reviews*（American Society for Microbiology）ほか

一般向け著書

一般読者・研究者・大学院生に向けた科学書・実用書を執筆。一覧は [著書ページ](#)。

2026

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2020

Aoki H., **Tani H.**, Nakamura K., Sato H., Torimura M., Nakazato T. MicroRNA biomarkers for chemical hazard screening identified by RNA deep sequencing analysis in mouse embryonic stem cells. *Toxicology and Applied Pharmacology* 392 (2020). doi:10.1016/j.taap.2020.114929

2019

Tani H., Numajiri A., Aoki M., Umemura T., Nakazato T. Short-lived long noncoding RNAs as surrogate indicators for chemical stress in HepG2 cells and their degradation by nuclear RNases. *Scientific Reports* 9 (2019). doi:10.1038/s41598-019-56869-y

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2018

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ノンコーディングRNAから生命の謎を解き明かす. (2018)

ヒトiPS細胞を用いた水質安全性評価のシステム. (2018)

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2017

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2016

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Mizutani R., Wakamatsu A., Tanaka N., Yoshida H., Tochigi N., Suzuki Y., Oonishi T., **Tani H.**, et al. Identification and Characterization of Novel Genotoxic Stress-Inducible Nuclear Long Noncoding RNAs in Mammalian Cells. *PLoS ONE* 7 (2012). doi:10.1371/journal.pone.0034949

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Fujimoto Y., Salam K.A., Furuta A., Matsuda Y., Fujita O., **Tani H.**, Ikeda M., Kato N., et al. Inhibition of Both Protease and Helicase Activities of Hepatitis C Virus NS3 by an Ethyl Acetate Extract of Marine Sponge Amphimedon sp. *PLoS ONE* 7 (2012). doi:10.1371/journal.pone.0048685

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Yamashita A., Salam K.A., Furuta A., Matsuda Y., Fujita O., **Tani H.**, Fujita Y., Fujimoto Y., et al. Inhibition of Hepatitis C Virus Replication and Viral Helicase by Ethyl Acetate Extract of the Marine Feather Star *Alloeocomatella polycladia*. *Marine Drugs* 10 (2012). doi:10.3390/md10040744

RNAの安定性を指標にして機能不明な長鎖ノンコーディングRNAの性質を定義する. (2012)

Up-frameshift protein 1 (UPF1): multitasking entertainer in RNA decay. *Drug discoveries & therapeutics* (2012)

2010

Miyata R., Adachi K., **Tani H.**, Kurata S., Nakamura K., Tsuneda S., Sekiguchi Y., Noda N. Quantitative detection of chloroethene-reductive bacteria *Dehalococcoides* spp. using alternately binding probe competitive polymerase chain reaction. *Molecular and Cellular Probes* 24 (2010). doi:10.1016/j.mcp.2009.11.005

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Noda N., **Tani H.**, Morita N., Kurata S., Nakamura K., Kanagawa T., Tsuneda S., Sekiguchi Y. Estimation of single-nucleotide polymorphism allele frequency by alternately binding probe competitive polymerase chain reaction. *Analytica Chimica Acta* 608 (2008). doi:10.1016/j.aca.2007.12.015

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Tani H., Kanagawa T., Kurata S., Teramura T., Nakamura K., Tsuneda S., Noda N. Quantitative Method for Specific Nucleic Acid Sequences Using Competitive Polymerase Chain Reaction with an Alternately Binding Probe. *Analytical Chemistry* 79 (2007). doi:10.1021/ac061506o

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2005

Tani H., Noda N., Yamada K., Kurata S., Tsuneda S., Hirata A., Kanagawa T. Quantification of Genetically Modified Soybean by Quenching Probe Polymerase Chain Reaction. *Journal of Agricultural and Food Chemistry* 53 (2005). doi:10.1021/jf048031r

谷を太字で示す。完全な最新リストは [研究業績ページ](#) / ORCID。